

Low Acrylonitrile NBR

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Abstract

The patent landscape for Low Acrylonitrile NBR (nitrile butadiene rubber) synthesis reveals a focus on emulsion polymerization processes. The majority of patents disclose the use of acrylonitrile, butadiene, and optionally methacrylic acid as monomers. Initiator systems, such as redox systems, are employed to facilitate polymerization. Emulsifier selection and loading are critical factors, with anionic surfactants being commonly used. The monomer ratio, particularly the acrylonitrile content, is carefully controlled to achieve desired properties. Coagulation and post-treatment conditions, including the use of coagulants and adjustment of total solids content, are also important considerations. The resulting Low Acrylonitrile NBR polymers exhibit a range of properties, including solid content, Mooney viscosity, and tensile properties. Overall, the patents demonstrate a trend towards developing Low Acrylonitrile NBR with improved performance and versatility.

Methodology

EP2316860 - Nitrile rubbers

- Process: emulsion polymerization
- Monomers: at least one alpha,beta-unsaturated nitrile, at least one conjugated diene and optionally at least one further copolymerizable monomer
- Monomer ratio: Not disclosed
- Initiator: redox system, which comprises (a) at least one peroxy compound as an oxidizing agent, (b) at least a reducing agent and (c) iron(II)chloride
- Emulsifier: Not disclosed
- Catalyst: Not disclosed
- Chain transfer agent: Not disclosed
- Coagulation: Not disclosed
- Water amount: Not disclosed

US9926439 - Nitrile rubber article having natural rubber characteristics

- Process: emulsion
- Monomers: acrylonitrile, butadiene, methacrylic acid

- Monomer ratio: acrylonitrile content between 17 and 45% by weight, methacrylic acid content less than 15% by weight, remaining balance butadiene
- Initiator: Not disclosed
- Emulsifier: anionic surfactant
- Catalyst: zinc oxide
- Chain transfer agent: Not disclosed
- Coagulation: Not disclosed
- Water amount: Not disclosed

EP2507313 - Elastomeric rubber and rubber products without the use of vulcanizing ...

- Process: emulsion
- Monomers: carboxylated acrylonitrile butadiene
- Monomer ratio: Not disclosed
- Initiator: Not disclosed
- Emulsifier: Not disclosed
- Catalyst: Not disclosed
- Chain transfer agent: Not disclosed
- Coagulation: coagulant dipping process
- Water amount: adjust to TSC 20-21%

Cross-Patent Comparison & Synthesis Trends

Emulsifier Selection and Loading

Anionic surfactants are commonly used as emulsifiers in Low Acrylonitrile NBR synthesis. However, the specific emulsifier selection and loading are not consistently disclosed across the patents.

Initiator and Chain Transfer Agent Strategies

Redox systems, comprising peroxo compounds, reducing agents, and iron(II)chloride, are employed as initiators. Chain transfer agents are not consistently disclosed.

Monomer Ratio and Reaction Control

The monomer ratio, particularly the acrylonitrile content, is critical in achieving desired properties. Acrylonitrile content is controlled between 17 and 45% by weight in some patents.

Coagulation and Post-Treatment Conditions

Coagulation conditions, including the use of coagulants and adjustment of total solids content, are important considerations. The coagulant dipping process is used in one patent, while others adjust the total solids content to specific ranges.

References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
EP2316860	Nitrile rubbers	Not disclosed	EP	2012	https://patents.google.com/patent/EP2316860
US9926439	Nitrile rubber article having natural rubber characteristics	Not disclosed	US	2018	https://patents.google.com/patent/US9926439
EP2507313	Elastomeric rubber and rubber products without the use of vulcanizing ...	Not disclosed	EP	2018	https://patents.google.com/patent/EP2507313